DESIGN AND IMPLEMENTATION OF AN IoT-BASED SMART NFT HYDROPONIC SYSTEM FOR LETTUCE AND ICEBERG CULTIVATION

Document scope and implementation status

This report preserves the principal subject matter and baseline design information in the supplied nine-page “Hydroponic Systems for Lettuce and Iceberg” document. It expands the design into an ESP32-based monitoring and control system using low-cost electronic project modules and the Blynk IoT platform.

Implementation-status statement: the electronic control and IoT subsystem is documented as the completed/assembled part of the project. The full-size NFT mechanical growing structure is not represented as a manufactured installation; it is a conceptual demonstration arrangement retained from the source document. Mechanical dimensions and hydraulic performance therefore require physical verification before crop production.

Trial-data statement: the ten-day dataset in Chapter 10 is generated illustrative data for demonstrating the reporting, trend analysis and dashboard functions. It is not presented as a set of measurements actually collected from the assembled hardware. Replace it with exported Blynk data or a laboratory log before submitting the report as experimental evidence.

Abstract

This project documents a low-cost Internet of Things (IoT) monitoring and control architecture for a Nutrient Film Technique (NFT) hydroponic system intended for lettuce and iceberg cultivation. An ESP32 development board acquires measurements from low-cost sensors for solution pH, estimated total dissolved solids (TDS), water temperature, ambient temperature and humidity, light intensity, reservoir level and nutrient-solution flow. Sensor readings are processed by firmware developed in the Arduino IDE and transmitted over Wi-Fi to Blynk Cloud. A mobile dashboard provides live parameter display, actuator status, manual relay control and historical trend visualization.

The proposed relay outputs switch a circulation pump, air pump, grow lights, cooling fan and reservoir refill pump. In automatic mode, firmware applies basic threshold and schedule logic while retaining safety interlocks. In manual mode, Blynk switches provide operator commands; protective interlocks remain active. Nutrient dosing and pH correction are intentionally manual because low-cost dosing without dedicated metering pumps and closed-loop calibration would introduce avoidable risk.

The report also defines sensor limitations, calibration procedures, wiring, Blynk datastreams, firmware, a ten-day illustrative dataset, system testing, failure modes and future improvements.

Keywords

Hydroponics; NFT; lettuce; ESP32; Arduino IDE; Blynk IoT; embedded systems; sensor monitoring; relay control; Internet of Things.

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1. Introduction and project objectives

1.1 Project background

Hydroponics is a horticultural method in which plants receive mineral nutrients through an aqueous solution rather than obtaining nutrients from field soil. The supplied source document identifies water efficiency, nutrient control, space efficiency, year-round cultivation, consistent produce quality and reduced soil-related disease pressure as major motivations for hydroponic lettuce production.

The selected configuration is Nutrient Film Technique (NFT). A shallow stream of nutrient solution is circulated through sloped channels, passes the plant root zone and returns by gravity to a reservoir. The electronic system monitors selected environmental and solution variables and provides local control plus remote visibility through a mobile application.

1.2 Problem statement

A conventional small NFT setup requires frequent manual observation. Operators must check solution condition, reservoir level, pump operation and the growing environment. A missed low-level condition or pump failure can interrupt nutrient delivery. This project addresses the monitoring and basic actuation requirements using affordable embedded hardware and an IoT dashboard.

1.3 Objectives

- Develop an ESP32-based embedded monitoring node using commonly available low-cost sensors.
- Measure solution pH, estimated TDS, water temperature, air temperature, relative humidity, light intensity, flow and reservoir level.
- Transmit telemetry to Blynk Cloud over Wi-Fi and display live values on a mobile dashboard.
- Provide a second dashboard page for historical graphs.
- Provide manual ON/OFF control and automatic control for five relay-operated loads.
- Implement safety interlocks for low reservoir level, high reservoir level and abnormal flow.
- Document wiring, firmware, calibration, commissioning and maintenance procedures.
- Present a ten-day illustrative dataset to demonstrate analysis of monitored variables.

1.4 Scope and exclusions

The scope includes embedded hardware, firmware, Wi-Fi communication, Blynk dashboards, relay switching and a conceptual mechanical NFT layout. The report does not claim that the mechanical channels, support table, reservoir plumbing or crop production trial were physically completed. Automated nutrient dosing, laboratory-grade water analysis, pH closed-loop dosing and commercial greenhouse climate control are excluded.

2. Source system and hydroponic background

2.1 Hydroponics and lettuce

The supplied document describes hydroponics as a method of growing plants in mineral nutrient solutions and notes historical developments in understanding plant nutrition. For lettuce, it emphasizes efficient water use, direct nutrient availability, space efficiency and repeatable growing conditions.

2.2 Types of hydroponic systems

System	Operating principle	Relevance to lettuce
Deep Water Culture (DWC)	Roots remain immersed in aerated nutrient solution.	Simple, but requires reliable aeration.
Drip irrigation	Nutrient solution is delivered to each plant through emitters.	More common for larger fruiting crops and substrate systems.
Aeroponics	Roots are suspended and periodically misted.	High control requirement; nozzle blockage is a concern.
NFT	A thin nutrient film flows through channels	Selected method; suitable for shallow-rooted

	and returns to a reservoir.	leafy greens.
Ebb and Flow	A tray is periodically flooded and drained.	Intermittent wetting; more complex timing.
Aquaponics	Fish waste is biologically converted into nutrients for plants.	Requires a balanced fish and plant ecosystem.

2.3 Why NFT was selected

NFT is suitable for lettuce because the root system can access a continuously renewed shallow nutrient stream while remaining exposed to air. The source document identifies nutrient delivery, oxygenation, water efficiency, consistent quality and ease of harvesting as advantages. NFT also has vulnerabilities: pump interruption, channel blockage, poor slope, temperature variation and reservoir contamination can rapidly affect root health.

2.4 Nutrient solution and source recommendations

The supplied document gives a target pH range of approximately 5.5–6.0 and discusses NPK, calcium and magnesium requirements. It mentions Engro Potash Power and a general 20-20-20 fertilizer as locally suggested materials, and gives an approximate starting recommendation of 1.5 g/L. These values are retained as source-document recommendations, not treated as a universally validated hydroponic recipe.

Engineering qualification: a general-purpose 20-20-20 fertilizer may not provide a complete hydroponic nutrient profile or the required balance of calcium, magnesium and micronutrients. Fertilizer composition varies by product. The operator must follow the product label and use a nutrient formulation suitable for hydroponics. Do not add fertilizer solely to force a TDS reading to a target. This project measures TDS as an approximate indicator; it does not identify individual ions.

2.5 Plant deficiencies

Nutrient	Symptoms summarized from source	Monitoring implication
Nitrogen	Reduced root growth; leaves become pale or yellow.	TDS alone cannot diagnose nitrogen deficiency.
Phosphorus	Dark green leaves; possible purple/red margins under prolonged deficiency.	Visual inspection remains necessary.
Potassium	Wilting, spotting and darkening leaf margins/tips.	Observe plant condition and nutrient formulation.
Calcium	Tip burn and distorted or wavy young leaves.	Maintain appropriate nutrient balance and environment.
Magnesium	Yellow/green discoloration near margins and leaf curling.	Check formulation; TDS does not identify magnesium.

2.6 Source mechanical arrangement

The source document lists a support table, PVC channels, substrate cups, reservoir, water pump and connecting hoses. It also mentions an air pump, grow lights and timers as additional equipment. It proposes a circulation pump flow rate of 0.75–1.0 L/min and a pump head of approximately 1.5 m. These figures are retained as preliminary design targets; the actual delivered flow depends on pump curve, tubing diameter, channel elevation and plumbing losses.

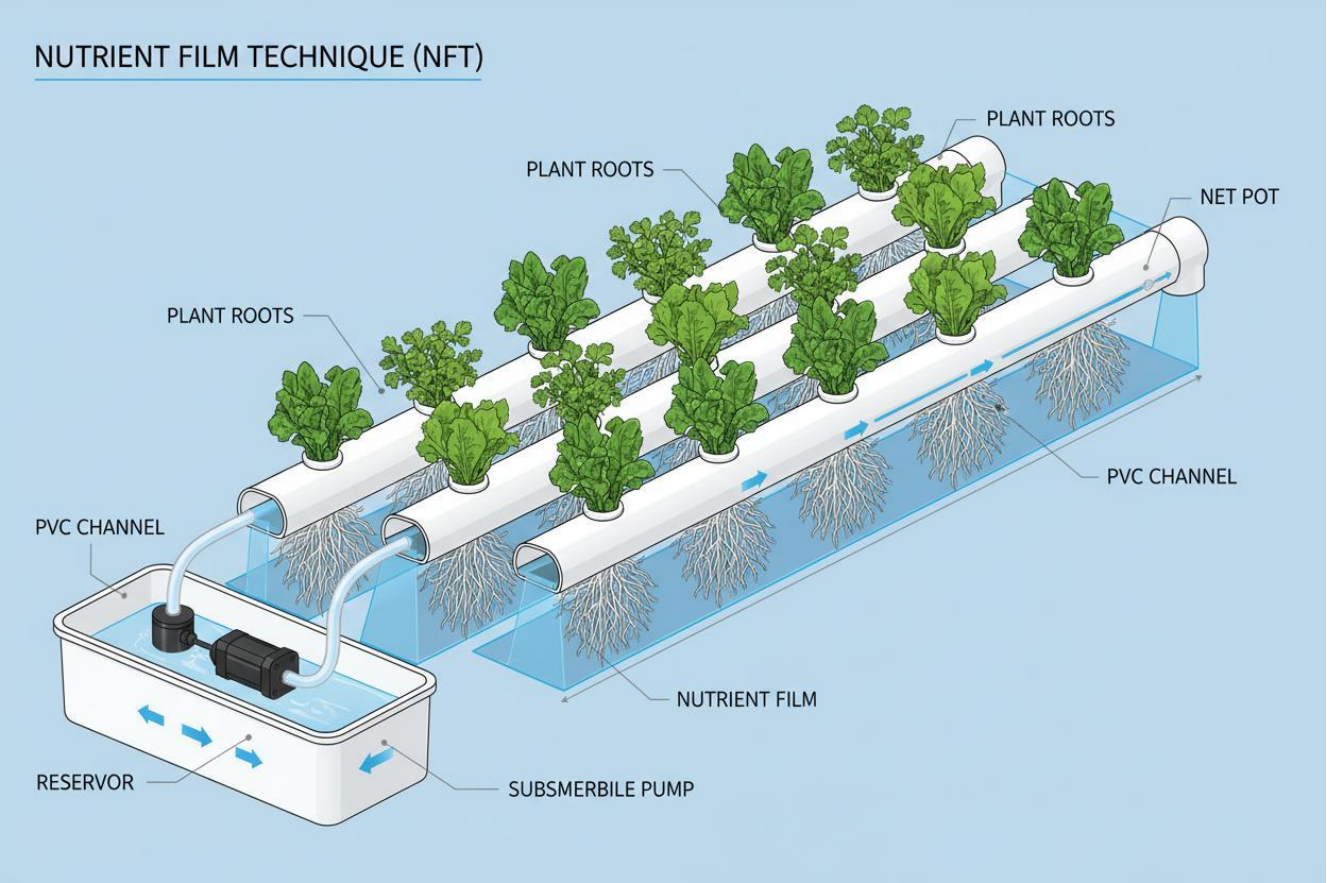


Figure 1. Conceptual NFT layout adapted to the source document. This is a demonstration schematic, not evidence of a manufactured mechanical structure.

2.7 Mechanical demonstration sequence

1. Mount the NFT channels on a rigid frame with a slight continuous slope toward the return outlet.
2. Drill planting holes at a suitable spacing and fit net pots or substrate cups.
3. Connect the reservoir outlet to the pump and pump discharge to the NFT channel inlet.
4. Route the channel outlet back to the reservoir by gravity.
5. Place the flow sensor inline where it remains flooded and accessible for cleaning.
6. Use opaque reservoir and tubing where possible to reduce light entry and algae growth.
7. Before planting, leak-test with clean water and verify that the flow reaches every channel without ponding.

3. System requirements and operating concept

3.1 Functional requirements

ID	Requirement	Implementation
FR-01	Measure solution pH and TDS estimate.	Analog pH and TDS interface modules.
FR-02	Measure water and ambient temperature.	Waterproof DS18B20 and DHT22.
FR-03	Measure humidity and light intensity.	DHT22 and BH1750.
FR-04	Monitor reservoir level.	Two float switches: low and high thresholds.
FR-05	Monitor nutrient flow.	YF-S401 low-flow Hall-effect sensor.
FR-06	Transmit readings to mobile dashboard.	ESP32 Wi-Fi and Blynk virtual datastreams.
FR-07	Show historical trends.	Blynk SuperChart / chart widget.
FR-08	Control five loads.	Relay board with manual and automatic modes.
FR-09	Protect pump from low-water operation.	Low-level interlock in firmware.

FR-10	Prevent refill overflow.	High-level interlock and refill timeout.

3.2 Overall system architecture

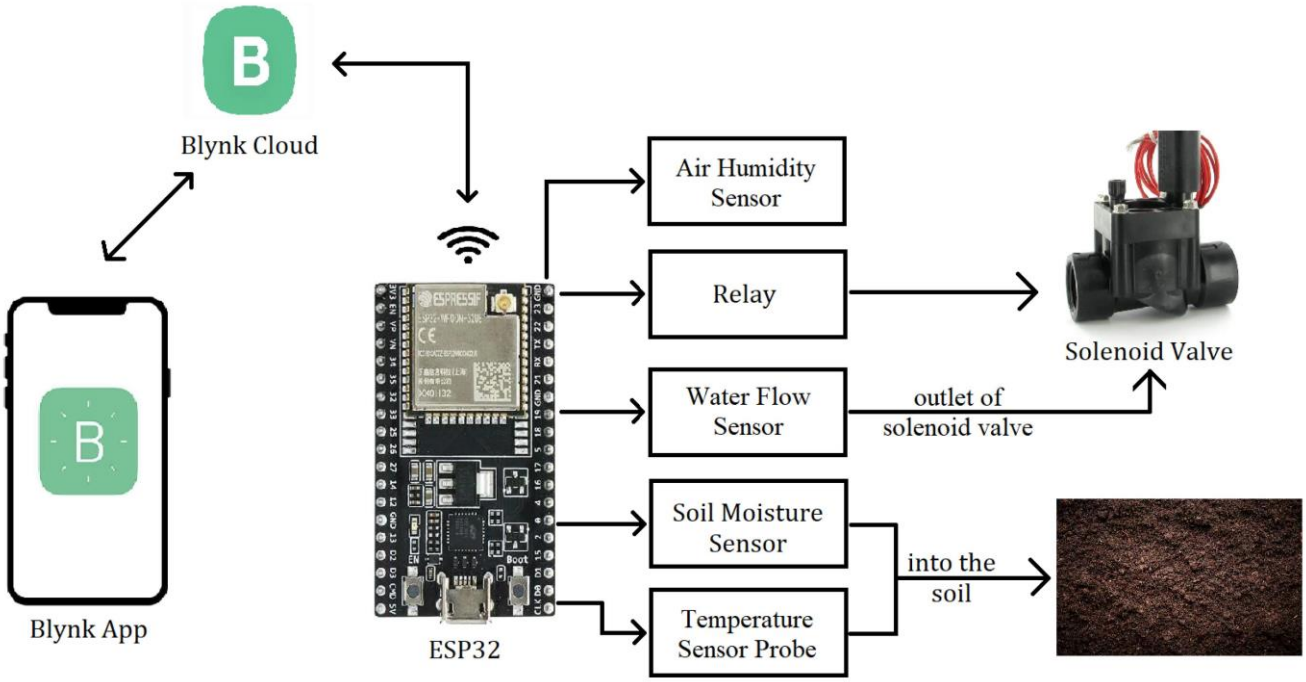


Figure 2. End-to-end IoT architecture: local sensing and control remain on the ESP32; Blynk provides remote monitoring and commands.

The ESP32 performs local control and does not depend on the cloud for basic automatic operation after Wi-Fi loss. Blynk is used for remote visibility, history and operator commands. The local control loop is deliberately simple and should be validated with the actual relay board and loads.

3.3 Operating modes

- AUTO: the ESP32 uses sensor values, reservoir switches and time schedule to control relays. The dashboard displays actual output states.
- MANUAL: the operator commands each relay through Blynk. Safety interlocks remain active; for example, the circulation pump cannot run when the low-level switch indicates insufficient water.
- Offline: if Wi-Fi or Blynk is unavailable, the ESP32 continues the local automatic rules. Remote manual commands are unavailable until reconnection.

3.4 Control thresholds

Function	Automatic rule used in this design	Engineering note
Circulation pump	ON when reservoir level is not LOW; OFF at LOW.	Normally continuous during operation; low-level protection has priority.
Air pump	ON when reservoir level is not LOW.	A simple relay output; no dissolved-oxygen sensor is included.
Grow lights	ON between 06:00 and 18:00 local time.	Schedule is configurable in firmware; actual crop photoperiod should be selected for the installation.
Cooling fan	ON above 28 °C; OFF below 26 °C.	Two thresholds provide hysteresis and reduce relay chatter.
Refill pump	ON at LOW level; OFF when HIGH level is reached.	90-second maximum run timer; operator must verify refill plumbing.
Flow alarm	Raise event if circulation pump is ON but flow remains below 0.15 L/min after a start delay.	Alarm threshold requires commissioning against real plumbing.
pH / TDS	Monitor and alert only; no automatic dosing.	Manual correction and re-measurement are required.

4. Embedded hardware design

4.1 Main controller

The selected controller is an ESP32 DevKit development board. It provides Wi-Fi connectivity, digital GPIO, interrupt inputs, ADC channels, I²C and 1-Wire interfaces. ADC1 pins are selected for analog measurements so that Wi-Fi operation does not conflict with ADC2 usage on classic ESP32 modules.

4.2 Hardware block diagram

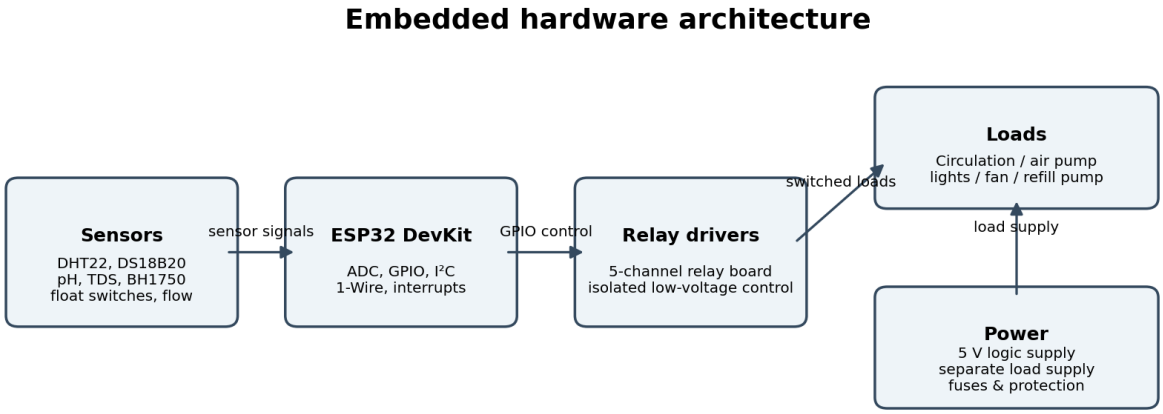


Figure 3. Embedded hardware blocks. The sensor supply, controller supply and actuator power paths must be designed separately and fused appropriately.

4.3 Bill of materials (indicative)

Item	Suggested low-cost module	Purpose / notes
1	ESP32 DevKit V1 (ESP-WROOM-32)	Main controller and Wi-Fi.
2	DHT22 / AM2302 module	Ambient temperature and relative humidity; lower-cost DHT11 is possible but less accurate.
3	Waterproof DS18B20 probe	Nutrient solution temperature; use 4.7 kΩ data pull-up.
4	Generic analog pH kit (PH-4502C-type interface + probe)	pH trend measurement; requires two-point calibration and probe storage solution.
5	Analog TDS sensor kit (Gravity-compatible or generic)	Approximate dissolved-solids indicator; temperature compensation and calibration required.
6	BH1750 digital light sensor	Illuminance in lux over I²C; lux is not equivalent to plant PPFD.
7	YF-S401 Hall-effect flow sensor	Low-flow range commonly specified around 0.3–6 L/min; confirm exact supplier variant and calibrate.
8	Two normally-open/closed float switches	Discrete low and high reservoir thresholds.
9	5-channel relay module	Switches circulation pump, air pump, light, fan and refill pump; verify 3.3 V input compatibility.
10	5 V regulated logic supply	Power relay board and suitable modules; ESP32 board via regulated USB/5 V input.
11	Separate load supply and fuse protection	Must match actual pump, fan and lighting voltage/current.
12	Resistors, terminal blocks, enclosure, wiring	Pull-up, level divider, connectors and electrical protection.

Indicative sensor choice is based on availability and cost, not on laboratory accuracy. Generic pH and TDS modules vary substantially between suppliers. The exact board pinout and output voltage must be checked before wiring.

4.4 Actuator outputs

Relay channel	Load	Control purpose
K1	Nutrient circulation pump	Recirculates solution through NFT channels.
K2	Air pump	Adds air to reservoir if the selected setup uses an air stone.
K3	LED grow lights	Provides scheduled lighting.
K4	Cooling fan	Moves air across the growing area.
K5	Reservoir refill pump	Adds clean water to maintain reservoir level; nutrient concentrate is not automatically dosed.

Relay contacts must be rated for the load type and inrush current, not merely the steady-state current. Mains-voltage switching must be enclosed, fused and performed with appropriate isolation and qualified supervision. The ESP32 and sensor wiring must remain physically separated from mains wiring.

4.5 Power architecture

Use a regulated 5 V supply for the relay board and modules that require 5 V. Power the ESP32 through its USB connector or a correctly regulated 5 V input. The analog sensor outputs connected to ESP32 ADC pins must never exceed the ESP32 input limit; use a divider or buffer where required. Do not connect a 5 V flow-sensor pulse output directly to a 3.3 V ESP32 GPIO. Use a resistor divider or logic-level interface. All low-voltage signal grounds must share a suitable reference, unless isolation is deliberately implemented.

4.6 Pin allocation

ESP32 pin	Connected device	Interface / notes
GPIO 4	DHT22 DATA	Digital; 10 kΩ pull-up to 3.3 V if module does not include one.
GPIO 5	DS18B20 DATA	1-Wire; 4.7 kΩ pull-up to 3.3 V.
GPIO 21	BH1750 SDA	I^2C data.
GPIO 22	BH1750 SCL	I^2C clock.
GPIO 34	pH analog output	ADC1 input-only; scale to safe voltage.
GPIO 35	TDS analog output	ADC1 input-only; scale to safe voltage.
GPIO 25	YF-S401 pulse output	Interrupt input; 5 V pulse must be level-shifted.
GPIO 26	High-level float switch	Digital input with pull-up.
GPIO 27	Low-level float switch	Digital input with pull-up.
GPIO 16	Relay K1	Circulation pump.
GPIO 17	Relay K2	Air pump.
GPIO 18	Relay K3	Grow lights.
GPIO 19	Relay K4	Cooling fan.
GPIO 23	Relay K5	Refill pump.

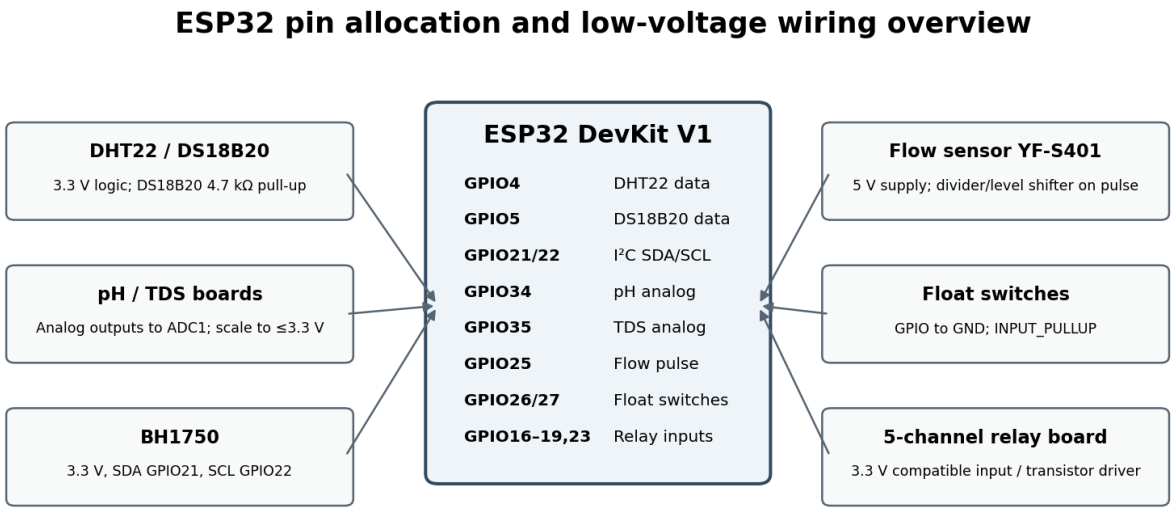


Figure 4. Wiring overview. The figure is a functional allocation diagram; follow the detailed wiring notes and verify each module's pinout before powering the circuit.

5. Sensor selection and measurement method

5.1 DHT22 ambient temperature and humidity

The DHT22 provides digital temperature and relative-humidity readings. It is inexpensive and widely used in student projects. It is not a precision environmental instrument and should be mounted away from direct water spray, hot power components and direct sunlight. A DHT11 can be substituted if availability is limited, with a corresponding reduction in accuracy and range.

5.2 DS18B20 water temperature

The waterproof DS18B20 probe measures reservoir solution temperature using a 1-Wire digital interface. A 4.7 kΩ pull-up resistor is normally connected between DATA and 3.3 V. The metal probe must be suitable for continuous water exposure and the cable entry should be protected from moisture.

5.3 Analog pH sensor

A low-cost analog pH interface board and glass probe provide a voltage related to solution pH. The relationship varies by board, probe condition, temperature and calibration. The firmware uses a configurable linear calibration model; the constants shown in the code are placeholders and must be replaced after calibration with pH 4.00 and pH 7.00 buffer solutions. A third buffer can be used to verify linearity.

- Rinse the probe with distilled/deionized water between buffers and samples; blot gently rather than rubbing the glass bulb.
- Do not store the pH probe dry. Use the storage solution specified by the probe supplier; do not store in distilled water unless the manufacturer explicitly instructs it.
- The low-cost interface output may exceed 3.3 V. Measure the output and scale/buffer it before connecting to GPIO34.

5.4 TDS sensor

The analog TDS module estimates total dissolved solids from electrical conductivity. It does not identify nutrient species and should not be interpreted as a complete nutrient analysis. Use a calibration solution of known conductivity/TDS,

account for temperature and record the conversion factor used. TDS values are often reported in ppm using a conversion factor from EC; different meters use different factors.

5.5 BH1750 light sensor

The BH1750 is a digital ambient light sensor connected over I^2C and reports illuminance in lux. It is suitable for a low-cost demonstration of changes in light level. Lux is weighted to human visual response and cannot be directly converted to photosynthetic photon flux density (PPFD) without a suitable spectral conversion and calibration. Mount the sensor near the plant canopy and avoid shading it with the enclosure.

5.6 Reservoir level switches

Two float switches provide discrete level detection. The lower switch identifies a LOW condition and the upper switch identifies a HIGH condition. This is less expensive and often more robust than attempting to measure a continuous percentage with a generic capacitive probe. The dashboard should therefore show LOW, NORMAL or HIGH, not a false precision percentage.

5.7 YF-S401 flow sensor

A YF-S401 Hall-effect flow sensor is selected because common variants are specified for approximately 0.3–6 L/min, which is closer to the source document's proposed 0.75–1.0 L/min than a typical YF-S201 flow meter. Supplier variants have different pulse constants; a commonly documented version is around 5,880 pulses/L. The firmware constant must be verified by collecting a measured volume over a timed interval. The flow sensor may add hydraulic restriction; confirm that the selected pump still delivers the required flow through the channels.

5.8 Sensor summary

Parameter	Sensor	Output	Calibration / limitation
Air temperature	DHT22	Digital °C	Compare with reference thermometer.
Relative humidity	DHT22	Digital %RH	Indicative; avoid condensation.
Water temperature	DS18B20	1-Wire °C	Check against reference thermometer.
pH	Analog pH probe + board	Analog voltage	Two-point pH buffer calibration.
TDS estimate	Analog TDS module	Analog voltage → ppm	Known standard; conversion factor varies.
Light	BH1750	I^2C lux	Lux is not PPFD.
Water level	2 float switches	Digital state	Verify switching height and polarity.
Flow	YF-S401	Pulse frequency	Bucket-and-stopwatch calibration; variant-specific K factor.

6. Electrical design and wiring

6.1 Low-voltage sensor wiring

8. Connect ESP32 3V3 to the DHT22 and DS18B20 logic supply where the module permits; connect grounds together.
9. Connect DHT22 DATA to GPIO4 and add a 10 kΩ pull-up to 3.3 V if required by the module.
10. Connect DS18B20 DATA to GPIO5 and use a 4.7 kΩ pull-up to 3.3 V.
11. Connect BH1750 SDA to GPIO21 and SCL to GPIO22. Use a 3.3 V-compatible module.
12. Connect pH and TDS analog outputs to GPIO34 and GPIO35 only after confirming and limiting their maximum voltage to the ESP32 ADC input range.
13. Power the YF-S401 at its specified supply voltage. Connect its pulse output to GPIO25 through a divider/level shifter if the output is 5 V.
14. Wire float switches to GPIO26 and GPIO27 with the other contact to GND. The firmware uses INPUT_PULLUP; confirm whether the physical switches are normally open or normally closed and adjust logic if needed.

6.2 Relay interface

Use a relay board with five independent channels and input compatibility with 3.3 V ESP32 logic. Some low-cost 5 V relay modules do not reliably recognize 3.3 V logic-high; test the exact board before final assembly. If necessary, use a transistor/ULN2003 driver stage or a dedicated logic-compatible driver. The example firmware assumes an active-LOW relay board, where LOW energizes a relay. Change RELAY_ON and RELAY_OFF if the selected board is active-HIGH.

6.3 Load-side switching

Each load is connected through the relay's COM and NO contacts so that the load is OFF when the relay is de-energized. Select the contact path, wire gauge, fuse and supply voltage based on the actual pump/fan/light ratings. For DC inductive loads, fit an appropriately rated flyback diode across the load if the load does not already contain suppression. AC loads require correctly rated snubbers/surge suppression where appropriate and must be enclosed.

6.4 Electrical safety

- Do not place exposed mains terminals on a breadboard or open prototype board.
- Use an insulated enclosure, strain relief, cable glands and correctly rated fuses.
- Keep water reservoirs and wet plumbing physically below and away from energized mains equipment.
- Use an RCD/GFCI-protected supply where applicable and maintain drip loops in cables.
- Never work on energized mains wiring. Have mains switching reviewed by a qualified person.
- Test all control logic first with low-voltage indicator lamps or dummy loads before connecting pumps or lighting.

7. Embedded software architecture

7.1 Development environment

Firmware is developed in Arduino IDE using the ESP32 board package and the following libraries: Blynk, DHT sensor library, OneWire, DallasTemperature and BH1750. Install the libraries through Arduino IDE Library Manager and select the correct ESP32 board and serial port.

7.2 Firmware tasks

Task	Operation
Sensor acquisition	Read temperature, humidity, pH, TDS, light, float switches and flow pulses.
Signal processing	Average analog samples and convert sensor signals using calibration constants.
Control logic	Apply AUTO/MANUAL selection, thresholds, time schedule and interlocks.
Blynk communication	Upload telemetry, receive switch commands and synchronize status.
Fault handling	Detect low reservoir, refill timeout and low flow while pump is commanded ON.
Watchdog / resilience	Avoid long blocking delays; allow local control to continue if cloud connection drops.

7.3 Analog conversion and calibration

ESP32 ADC readings are affected by attenuation, chip variation, input impedance and noise. Use analogReadMilliVolts where supported by the selected Arduino-ESP32 core, average several samples and calibrate the complete sensor-plus-interface chain. Never assume the raw ADC count maps linearly to pH or TDS without calibration.

7.4 Automatic mode logic

Automatic rules are implemented locally on the ESP32. The circulation and air pumps are enabled when the reservoir is above the low-level switch. The refill pump runs when the low-level switch indicates LOW and stops when the high-level switch indicates HIGH or when the maximum refill time expires. Grow lights follow the local-time schedule. The fan uses a 28 °C ON and 26 °C OFF hysteresis band. Flow is monitored to identify a likely pump, tubing or blockage problem.

7.5 Manual mode logic

In manual mode, each Blynk switch updates a virtual-pin command. The firmware changes the associated relay only after applying safety interlocks. Manual control does not bypass low-level pump protection or the high-level refill cutoff. Actual relay status is written back to Blynk so the dashboard reflects the output state rather than merely the requested command.

7.6 Network-loss behavior

If Wi-Fi or Blynk Cloud becomes unavailable, the local automatic control continues using the most recent sensor readings. The operator cannot issue remote commands during the outage. On reconnection, the device synchronizes the mode and command datastreams. For a real installation, use a defined safe startup state and verify that no unexpected relay energization occurs after reset.

8. Blynk IoT dashboard design

8.1 Platform structure

Create a Blynk template for ESP32 over Wi-Fi, create virtual-pin datastreams, add a device from the template and configure the mobile dashboard. Blynk documentation distinguishes the mobile dashboard from the web dashboard; both can use the same datastreams. The mobile app dashboard is configured separately.

8.2 Page 1 — live monitoring and manual control

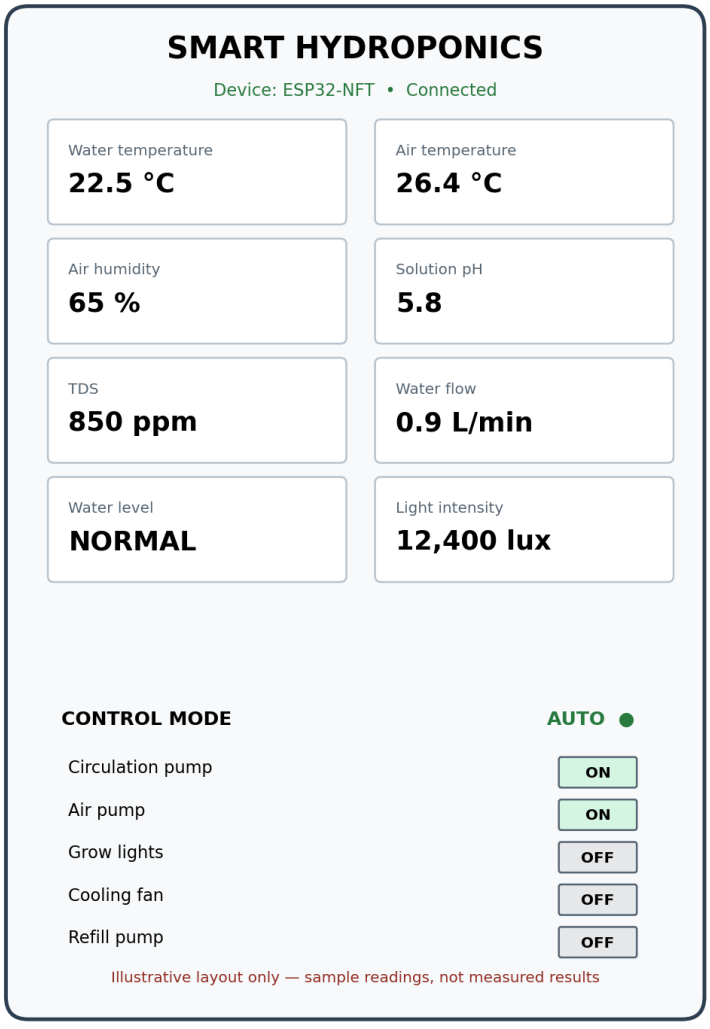


Figure 5. Proposed Blynk mobile live dashboard mock-up. Readings shown are illustrative interface values, not measurements.

Page 1 includes the eight live data items: water temperature, air temperature, air humidity, solution pH, TDS, water flow, reservoir level and light intensity. It also includes the AUTO/MANUAL mode selector and five ON/OFF controls. The mode selector should be clearly visible; manual switches should be disabled or visually marked as inactive while AUTO mode is selected.

8.3 Page 2 — historical graphs

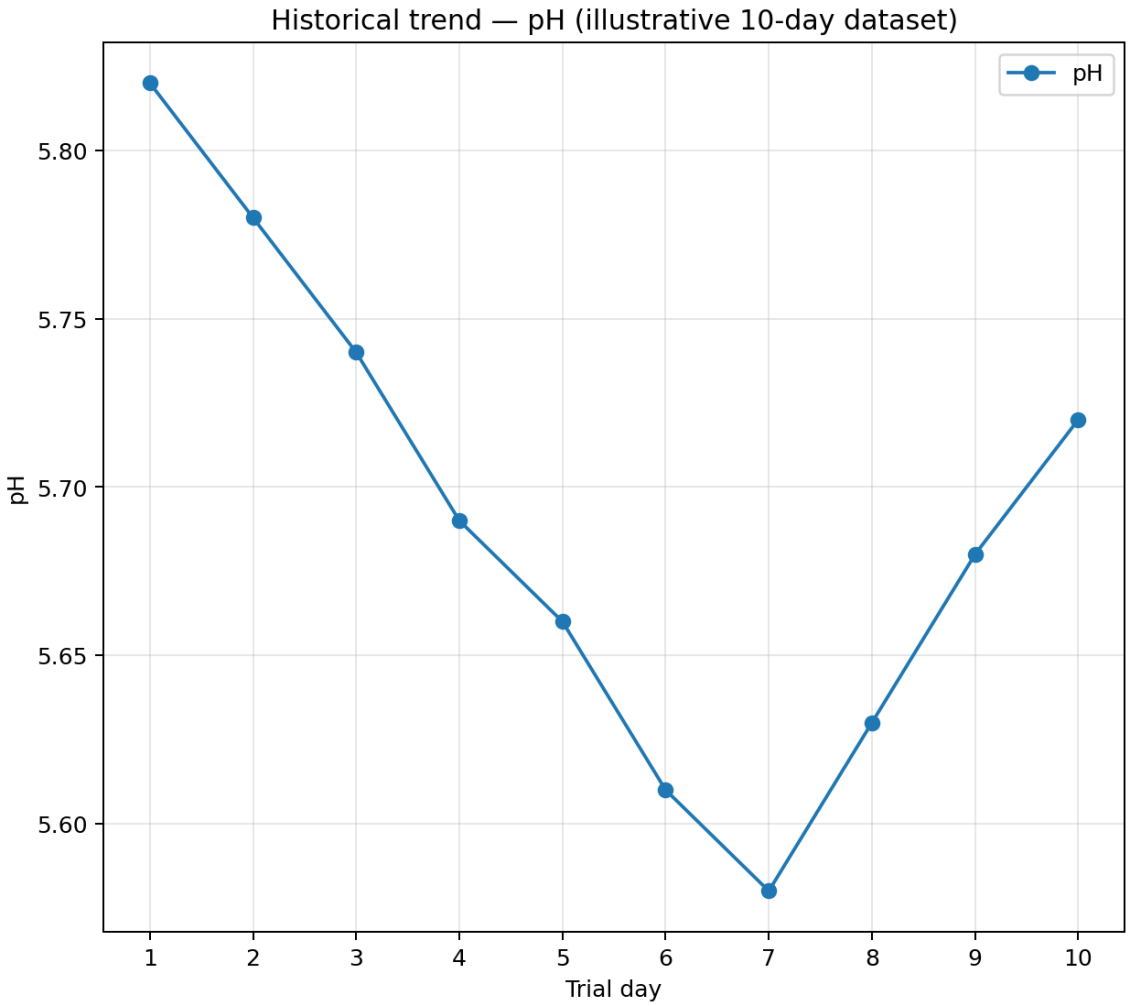


Figure 6. Example historical pH trend using the illustrative ten-day dataset. In Blynk, configure a SuperChart with the sensor datastreams.

The second dashboard page is dedicated to historical trends. Configure a SuperChart for pH, TDS, water temperature, air temperature, humidity, flow and light intensity. Because the parameters use different units and ranges, avoid plotting all values on one unscaled axis. Use separate chart widgets or group compatible variables. Enable time-range selection and a visible legend.

8.4 Virtual-pin and datastream map

Virtual pin	Datastream	Direction	Data type / range
V0	pH	ESP32 → Blynk	Double; 0–14
V1	TDS_ppm	ESP32 → Blynk	Double; 0–3000
V2	WaterTemp_C	ESP32 → Blynk	Double; 0–50
V3	AirTemp_C	ESP32 → Blynk	Double; 0–60
V4	Humidity_RH	ESP32 → Blynk	Double; 0–100
V5	WaterLevel_State	ESP32 → Blynk	Integer; 0 LOW, 1 NORMAL, 2 HIGH
V6	Flow_Lmin	ESP32 → Blynk	Double; 0–6

V7	Light_lux	ESP32 → Blynk	Integer; 0–65535
V8	AutoMode	App → ESP32	Integer; 0 MANUAL, 1 AUTO
V9	Cmd_Circulation	App → ESP32	Integer; 0/1
V10	Cmd_AirPump	App → ESP32	Integer; 0/1
V11	Cmd_GrowLight	App → ESP32	Integer; 0/1
V12	Cmd_Fan	App → ESP32	Integer; 0/1
V13	Cmd_Refill	App → ESP32	Integer; 0/1
V14	Actual_Circulation	ESP32 → App	Integer; 0/1
V15	Actual_AirPump	ESP32 → App	Integer; 0/1
V16	Actual_GrowLight	ESP32 → App	Integer; 0/1
V17	Actual_Fan	ESP32 → App	Integer; 0/1
V18	Actual_Refill	ESP32 → App	Integer; 0/1
V19	System_Alert	ESP32 → App	Integer; 0 normal, 1 alert

8.5 Dashboard setup steps

15. Create a Blynk account and create a new template named Smart NFT Hydroponics; choose ESP32 and Wi-Fi.
16. Create the datastreams using the virtual-pin map above. Set units, minimum and maximum values, and numeric data types.
17. Create a device from the template and copy its Template ID, Template Name and Auth Token into the firmware.
18. Configure the mobile dashboard live page with labeled values/gauges for V0–V7, a mode switch on V8 and five actuator switches on V9–V13.
19. Add a second dashboard page or separate dashboard view containing chart widgets bound to V0–V7. Blynk mobile and web widgets are configured independently.
20. Create Blynk events named flow_fault, low_reservoir and refill_timeout if event notifications are desired; match the names in firmware.
21. Upload firmware, connect the device and verify each datastream updates before connecting actual loads.

9. Assembly, configuration and commissioning

9.1 Hardware assembly sequence

22. Mount the ESP32, sensor interface boards and relay board in a ventilated, insulated electronics enclosure.
23. Wire the sensors to the specified GPIO pins and confirm all signal voltages with a multimeter.
24. Connect the relay board inputs and verify active-LOW or active-HIGH behavior using indicator LEDs or low-voltage dummy loads.
25. Install the pH and TDS probes in the reservoir sample zone without allowing cables or connectors to become submerged.
26. Install the DS18B20 probe in the water and locate the DHT22 away from direct spray.
27. Install the float switches at the low and high reservoir levels and test the switch transitions manually.
28. Install the flow sensor in the intended direction and ensure the tubing connection does not leak.
29. Connect the ESP32 to Wi-Fi and Blynk; verify sensor values and command acknowledgements.
30. Test automatic rules and safety interlocks with dummy loads before operating pumps or lights.

9.2 Arduino IDE setup

31. Install Arduino IDE and the Espressif ESP32 board package.
32. Select the ESP32 Dev Module board and the correct USB serial port.
33. Install Blynk, DHT sensor library, OneWire, DallasTemperature and BH1750 libraries.
34. Open the firmware in Appendix A and enter Wi-Fi SSID, password, Blynk Template ID, Template Name and Auth Token.
35. Set the sensor calibration constants after calibration; confirm relay polarity and flow pulse constant.
36. Compile and upload; open Serial Monitor at 115200 baud for diagnostics.
37. Create the Blynk device and confirm the mobile dashboard values match serial output.

9.3 Calibration procedures

Sensor	Calibration procedure	Acceptance check
pH	Use pH 7.00 and pH 4.00 buffers; record voltage at each point and calculate slope/intercept.	Displayed value within the probe/module's realistic tolerance; repeat after rinsing.
TDS	Use a known conductivity/TDS standard; record voltage and adjust conversion factor.	Reading is repeatable and close to standard at reference temperature.
DS18B20	Compare in a stirred water bath with a reference thermometer.	Difference is documented; investigate if unstable.
DHT22	Compare against a reference thermo-hygrometer under stable conditions.	Stable readings; no condensation.
BH1750	Compare relative change against a known lux meter if available.	Consistent response to changing illumination.
Float switches	Raise/lower water level across both switch heights.	LOW/NORMAL/HIGH states correspond to physical positions.
Flow sensor	Collect a measured volume for a measured time; calculate L/min and pulse constant.	Displayed flow agrees within a documented tolerance.

9.4 Flow calibration calculation

If a sensor produces N pulses during a measurement interval t seconds and the calibrated constant is K pulses per litre, then volume V = N/K litres and flow Q = 60V/t L/min. Use a measured bucket volume and stopwatch to determine K for the exact sensor variant. The example code starts with 5,880 pulses/L, but this must not be assumed correct for every YF-S401 product.

10. Ten-day illustrative monitoring trial

10.1 Trial protocol

The following ten-day series is a generated demonstration dataset intended to show how a Blynk monitoring trial could be tabulated and analyzed. It is not a record exported from the constructed hardware. The values are plausible illustrative averages selected to demonstrate small day-to-day changes, sensor drift and stable operation. The report must not describe these values as measured experimental observations.

Assumed demonstration conditions: NFT circulation target 0.75–1.0 L/min; solution pH near the source document's 5.5–6.0 range; reservoir level maintained in NORMAL state; lights scheduled during the day; air pump enabled; fan threshold set at 28 °C. pH and TDS are monitoring-only variables.

10.2 Daily dataset

Day	Water °C	Air °C	RH %	pH	TDS ppm	Flow L/min	Light lux	Level	Main	Air	Light
Day 1	22.1	25.2	64	5.82	820	0.91	12,500	Normal	ON	ON	ON
Day 2	22.4	25.8	63	5.78	828	0.90	12,700	Normal	ON	ON	ON
Day 3	22.7	26.1	61	5.74	835	0.89	12,400	Normal	ON	ON	ON
Day 4	23.0	27.0	60	5.69	842	0.88	12,100	Normal	ON	ON	ON
Day 5	22.8	27.8	58	5.66	850	0.87	11,800	Normal	ON	ON	ON
Day 6	23.2	28.4	57	5.61	858	0.86	11,600	Normal	ON	ON	ON
Day 7	23.5	27.6	59	5.58	866	0.85	11,900	Normal	ON	ON	ON
Day 8	23.1	26.9	62	5.63	852	0.88	12,300	Normal	ON	ON	ON
Day 9	22.9	27.2	61	5.68	844	0.90	12,600	Normal	ON	ON	ON
Day 10	23.3	26.5	64	5.72	838	0.89	12,400	Normal	ON	ON	ON

10.3 Trend graphs

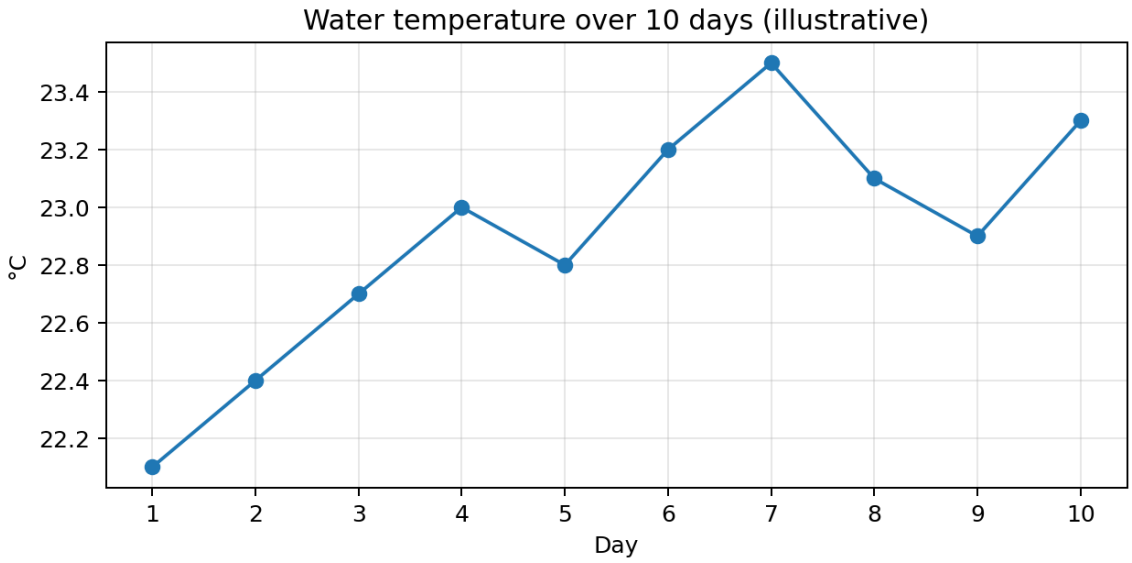


Figure 7. Illustrative water-temperature trend.

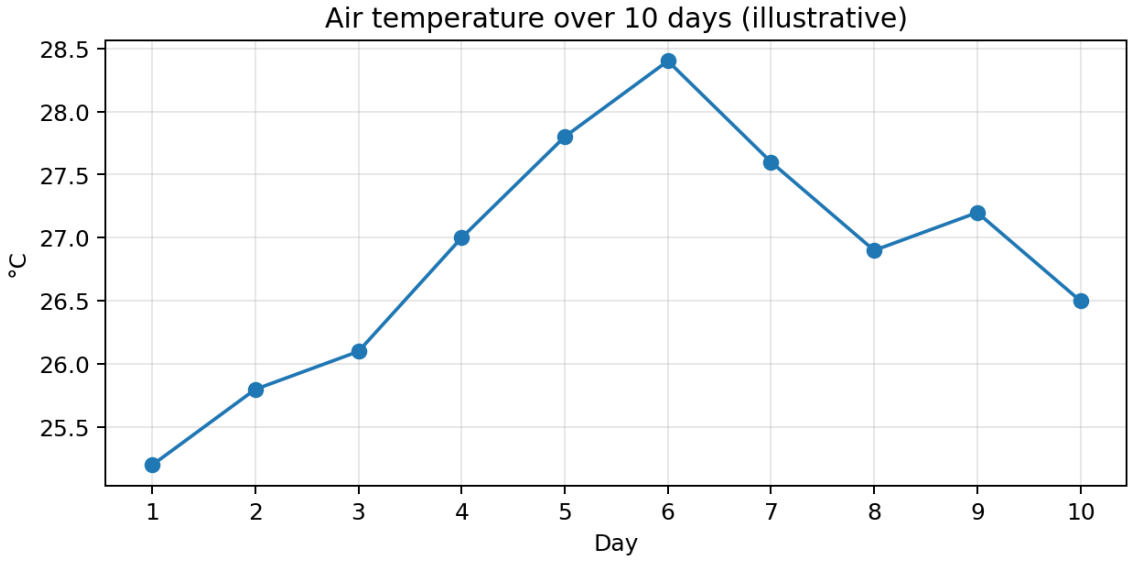


Figure 8. Illustrative ambient-temperature trend.

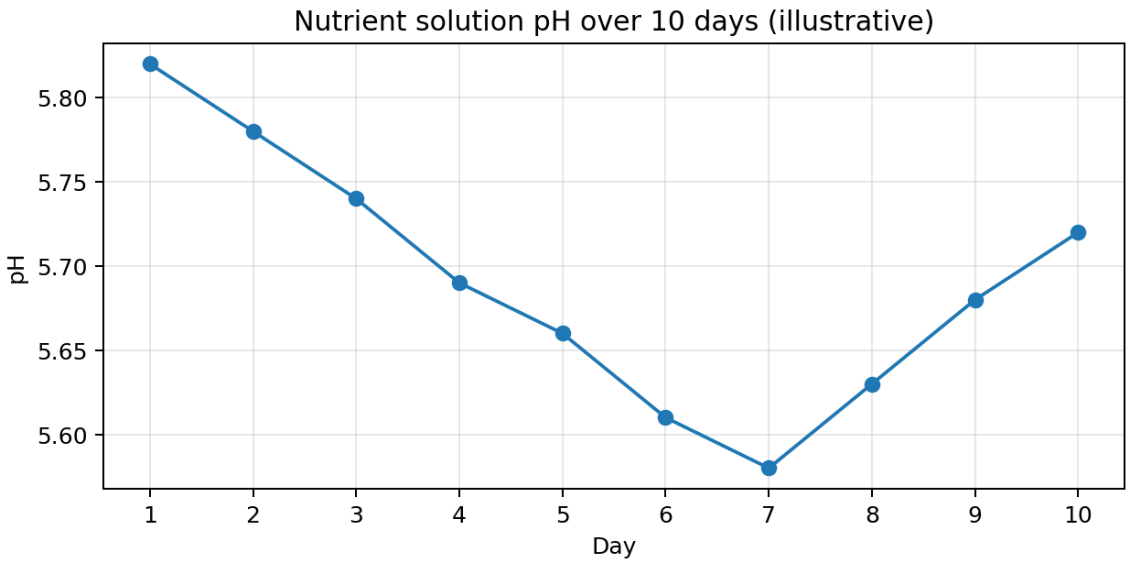


Figure 9. Illustrative pH trend.

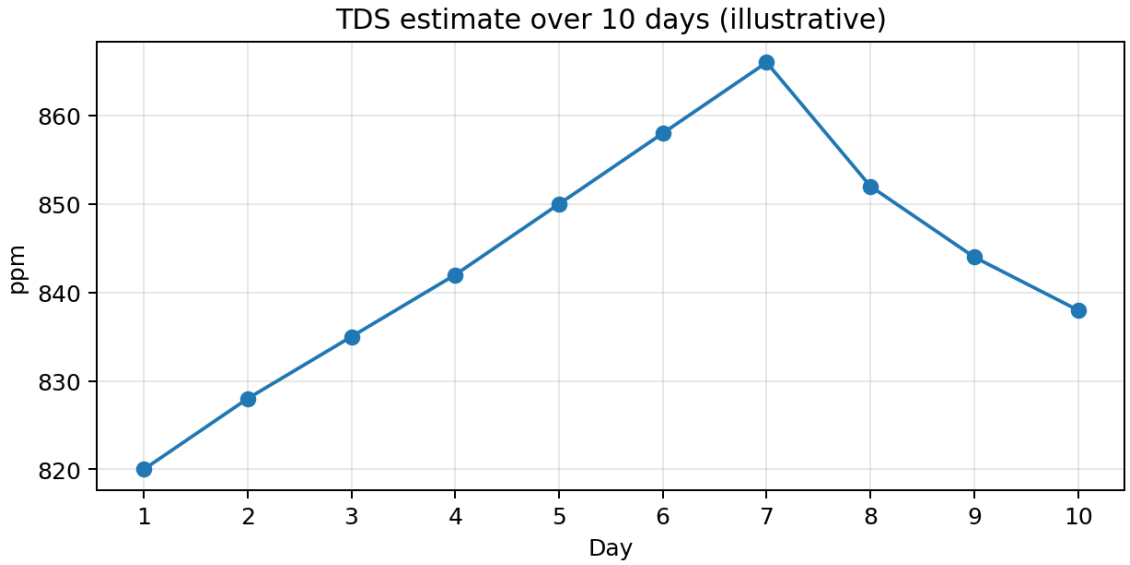


Figure 10. Illustrative TDS trend.

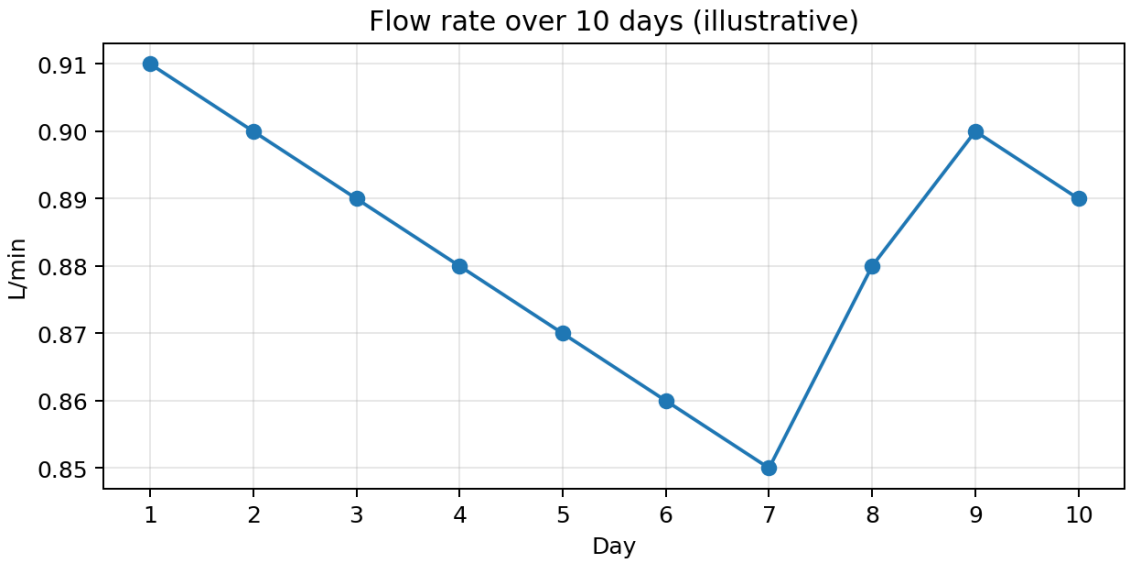


Figure 11. Illustrative flow trend.

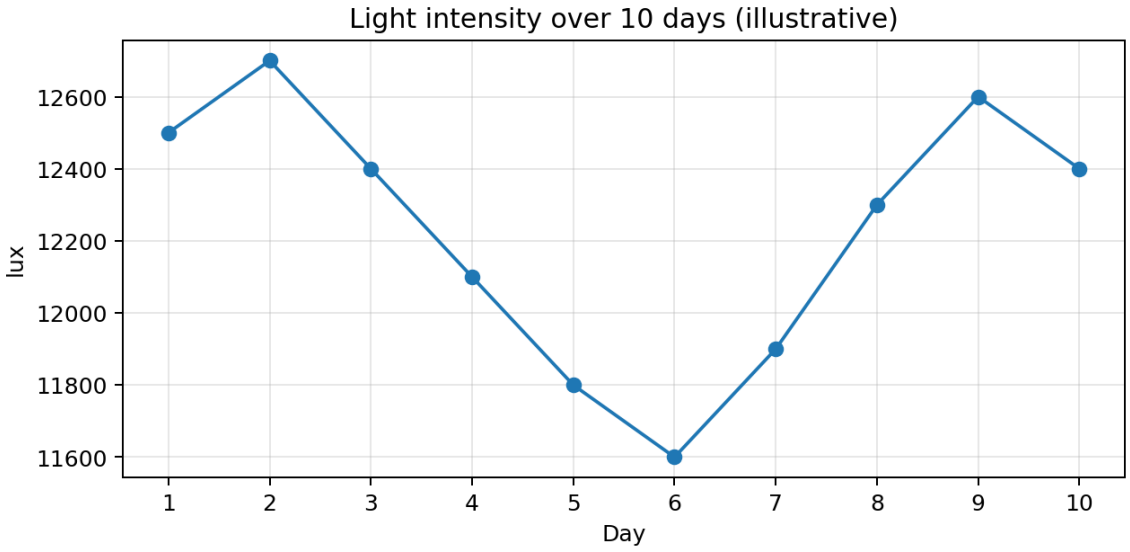


Figure 12. Illustrative light-intensity trend.

10.4 Summary statistics

Parameter	Illustrative range	Illustrative mean
Water temperature	22.1–23.5 °C	22.90 °C
Air temperature	25.2–28.4 °C	26.85 °C
Relative humidity	57–64 %	60.9 %
pH	5.58–5.82	5.69
TDS	820–866 ppm	843.3 ppm
Flow rate	0.85–0.91 L/min	0.88 L/min
Light intensity	11,600–12,700 lux	12,230 lux

10.5 Interpretation

The illustrative pH series decreases gradually from 5.82 to 5.58 before recovering to 5.72. This demonstrates why a trend view is useful: a single reading may appear acceptable while a gradual drift indicates that the operator should verify calibration and solution condition. No automatic pH correction is assumed.

The TDS estimate rises from 820 ppm to 866 ppm and then decreases to 838 ppm. This may represent concentration changes from water loss, plant uptake, top-up water or measurement variation. A TDS meter cannot identify which dissolved ions changed; a reading alone is not sufficient to decide which fertilizer to add.

The water-temperature series remains between 22.1 and 23.5 °C in this example. Air temperature briefly exceeds the 28 °C fan threshold on Day 6, illustrating an automatic fan command. The flow series remains close to the proposed 0.75–1.0 L/min design target and gradually recovers after a small decrease. Actual values would depend on the pump curve, channel height, tubing and flow sensor calibration.

The reservoir level is shown as NORMAL throughout the ten days and the circulation pump, air pump and lights are shown ON in the daily summary. The fan and refill pump are not included in the compact table because their ON/OFF duration would depend on the sampling interval; in a real trial, log relay state changes and runtime counters as separate datastreams.

10.6 What to log in a real trial

For a defensible experimental trial, export timestamped Blynk data or record readings at a fixed interval (for example, every five minutes). Include sensor calibration records, time zone, sensor placement, water top-ups, nutrient additions, cleaning, plant observations, relay state changes, Wi-Fi interruptions and any missing data. Use actual data in place of the illustrative series before claiming experimental performance.

11. Results and engineering discussion

11.1 Expected system behavior

The design demonstrates an integrated embedded system with sensing, local decision-making, remote telemetry and actuator control. Its main engineering value is not high-precision nutrient chemistry; it is the combination of affordable sensing, trend visualization, simple automation and basic equipment protection.

11.2 Accuracy and uncertainty

- pH accuracy depends on the probe quality, calibration buffers, analog interface, temperature and electrical noise.
- TDS is an estimated aggregate measurement and does not reveal nutrient composition.
- DHT22 measurements are suitable for basic environmental monitoring but not precision greenhouse control.
- BH1750 measures lux, not plant-usable photon flux directly.
- Float switches provide discrete levels rather than a continuous percentage.
- Flow-sensor pulse constants differ between manufacturers and variants; calibrate the exact unit.

11.3 Engineering trade-offs

The low-cost sensor strategy reduces project cost and makes the system reproducible for a student laboratory. The trade-off is increased calibration effort and reduced traceability compared with industrial probes. The design therefore uses pH and TDS primarily as trend indicators, preserves manual nutrient correction and uses simple float switches for reliable safety states.

11.4 Verification matrix

Test	Method	Pass condition
Sensor acquisition	Compare serial output with sensor response and reference instruments where available.	All channels return plausible values without blocking firmware.
Blynk telemetry	Observe dashboard while changing sensor inputs.	Datastreams update and labels/units are correct.
Manual relay control	Switch each output from mobile dashboard in MANUAL mode.	Correct relay operates; status is returned to app.

AUTO mode	Simulate threshold crossings and time schedule.	Relays follow documented rules.
Low-level interlock	Activate LOW float switch.	Circulation and air pump outputs turn OFF; refill logic may start.
High-level cutoff	Activate HIGH float switch.	Refill pump turns OFF.
Flow fault	Stop/obstruct flow during safe dummy-load test.	Flow fault event is generated after delay.
Wi-Fi loss	Disconnect access point during AUTO operation.	Local automatic control continues; dashboard reconnects later.

12. Safety, limitations and maintenance

12.1 Known limitations

- The mechanical NFT structure is a demonstration design and has not been manufactured or hydraulically validated.
- The illustrative ten-day values are generated for documentation and are not experimental data.
- The low-cost pH and TDS modules require calibration and may drift.
- The project does not measure dissolved oxygen, electrical conductivity with a laboratory instrument, nutrient ions or water microbiology.
- The automatic rules are basic threshold control, not an agronomic optimization algorithm.
- Cloud dashboard access depends on Wi-Fi and Blynk service availability.
- The refill pump logic requires correctly installed float switches and a reliable clean-water source.

12.2 Maintenance schedule

Frequency	Maintenance action
Daily	Check reservoir level, flow, leaks, pump operation, plant appearance and dashboard connectivity.
Every 2–3 days	Inspect channels for root blockage and algae; verify pH/TDS readings are plausible.
Weekly	Rinse probes as recommended, inspect float switches, clean filter and verify pump flow.
Before each trial	Calibrate pH and TDS sensors; test all relay interlocks and alarms.
Monthly	Inspect wiring, terminals, enclosure seals, relay contacts and power supplies.
As required	Replace damaged probes, tubing, pump or relay module; re-run commissioning tests.

12.3 Failure modes and mitigations

Failure mode	Effect	Mitigation
Circulation pump failure	Nutrient film stops flowing.	Flow alarm; inspect pump, power and blockage.
Reservoir low	Pump may run dry; roots may dry.	Low-level float interlock; refill and investigate leaks.
Refill switch stuck	Potential overflow or failure to refill.	High-level cutoff, maximum runtime and manual inspection.
pH probe drift	Incorrect pH display.	Two-point calibration; compare with buffer and replace aged probe.
TDS sensor fouling	Incorrect nutrient estimate.	Clean and recalibrate; avoid interpreting as ion-specific data.
Wi-Fi/cloud loss	Remote view/control unavailable.	Local control continues; reconnect and review logs.
Relay contact failure	Load remains ON or OFF unexpectedly.	Fuse loads; test contact operation; replace defective relay.
Water ingress	Electrical short or corrosion.	Enclosure, drip loops, cable glands and separation from water.

13. Conclusion and future work

A low-cost IoT architecture for an NFT hydroponic lettuce system has been defined using an ESP32, Arduino IDE, common sensor modules, relay outputs and Blynk Cloud. The design monitors solution and environmental variables, provides live and historical dashboards, and supports both automatic and manual control. The automatic control layer includes basic pump interlocks, reservoir refill logic, a lighting schedule, fan hysteresis and flow-fault detection.

The report distinguishes the completed electronic/IoT subsystem from the unmanufactured demonstration mechanical structure. It also distinguishes generated illustrative trial values from actual experimental evidence. Before a production crop trial, the mechanical NFT assembly must be built and leak-tested, all sensors calibrated, relay and mains wiring reviewed, and the firmware verified under fault conditions.

13.1 Recommended future improvements

- Replace generic pH/TDS boards with better characterized probes and isolated signal conditioning if the budget allows.
- Add a proper EC sensor with temperature compensation and periodic calibration.
- Add a dissolved-oxygen sensor only if budget and maintenance capacity justify it.
- Add a real-time clock or robust network time handling for lighting schedules.
- Add relay runtime counters, pump energy measurement and a local buzzer/display.
- Use a watchdog, brownout-safe relay defaults and a hardware emergency stop.
- Add a calibrated PPFD sensor if lighting optimization is a project objective.
- Build the mechanical NFT structure and perform a measured flow-distribution test across every channel.